

The *Mycobacterium tuberculosis* complex pangenome is small and shaped by sub-lineage-specific regions of difference

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eLife Assessment

This **useful** study analyzed 335 *Mycobacterium tuberculosis* Complex genomes and found that MTBC has a closed pangenome with few accessory genes. The research provides **solid** evidence for gene presence-absence patterns which support the appending conclusions however, the main criticism regarding the dominance of genome reduction remains.

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Abstract

The *Mycobacterium tuberculosis* complex (MTBC) is a group of bacteria causing tuberculosis (TB) in humans and animals. Understanding MTBC genetic diversity is crucial for insights into its adaptation and traits related to survival, virulence, and antibiotic resistance. While it is known that within-MTBC diversity is characterised by large deletions found only in certain lineages (regions of difference [RDs]), a comprehensive pangenomic analysis incorporating both coding and non-coding regions remains unexplored. We utilised a curated dataset representing various MTBC genomes, including under-represented lineages, to quantify the full diversity of the MTBC pangenome. The MTBC was found to have a small, closed pangenome with distinct genomic features and RDs both between lineages (as previously known) and between sub-lineages. The accessory genome was identified to be a product of genome reduction, showing both divergent and convergent deletions. This variation has implications for traits like virulence, drug resistance, and metabolism. The study provides a comprehensive understanding of the MTBC pangenome, highlighting the importance of

genome reduction in its evolution, and underlines the significance of genomic variations in determining the pathogenic traits of different MTBC lineages.

Introduction

The *Mycobacterium tuberculosis* complex (MTBC) is the term used for *M. tuberculosis* and all its variants, many of which used to be separate species (Riojas et al., 2018). These bacteria are the primary cause of tuberculosis (TB) in humans and other animals. Worldwide, TB is the 13th leading cause of death and is once again the top infectious killer following global efforts to combat the COVID-19 pandemic (WHO, 2024). The MTBC comprises both human-associated lineages (L1 to L10) and animal-associated lineages (La1, La2, La3, *M. tuberculosis* var *microti* and *M. tuberculosis* var *pinnipedi*) (Coscolla et al., 2021, Zwyrer et al., 2021, Guyeux et al., 2024). The human-adapted MTBC exhibits a strong phylogeographical population structure, with some lineages occurring globally and others showing a strong geographical restriction (Gagneux, 2018). For example, L2 and L4 are widespread globally, with L2 dominating in Southeast Asia and L4 in Europe (Napier et al., 2020). L1 and L3 occur mainly in regions around the Indian Ocean (Menardo et al., 2021), L5 and L6 are highly restricted to West Africa (de Jong et al., 2010) and L7-L10 are exclusively found in Central and East Africa (Blouin et al., 2012, Ngabonziza et al., 2020, Coscolla et al., 2021, Guyeux et al., 2024). Animal-associated lineages refer to lineages of the MTBC that primarily infect and cause disease in animals but can also infect humans in close contact with infected animals (Brites et al., 2018). Little is known about the geographic distribution and host range of animal-associated lineages (Zwyrer et al., 2021).

Various factors such as human migration, local adaptation to host population, and host immune response are thought to have contributed to the current phylogeographical distribution of human-adapted MTBC members (Darwin and Stanley, 2022, Hershberg et al., 2008). Additionally, different lineages of MTBC are known to differ in virulence, metabolism and transmission potential, which can affect the spread of certain lineages and sub-lineages (Coscolla, 2017, Gagneux et al., 2006). Understanding the genetic diversity within MTBC is essential for gaining insights into how these bacteria adapt to different host populations and environments, and identify traits that may be important for survival, virulence, and antibiotic resistance.

MTBC is recognised for its clonal population structure and little genetic diversity between different strains, caused by the absence of horizontal gene transfer and plasmids (Godfroid et al., 2018, Chiner-Oms et al., 2022). Single nucleotide polymorphisms (SNPs) and large sequence polymorphisms (LSPs) represent the source of MTBC genetic diversity (Gagneux, 2018). Within the MTBC, LSPs which are deletions in genomes relative to the reference genome H37Rv are referred to as Regions of Difference (RDs) (Bespiatykh et al., 2021). These RDs are known to differentiate MTBC lineages from each other (Gagneux, 2018), but within-lineage diversity is thought to be mostly composed of SNPs (Napier et al., 2020). While some RDs have been described to be differentially present within some lineages (Sanoussi et al., 2021, Bespiatykh et al., 2021), these are thought to be rare and remain unexplored. RDs can surpass 10 kb in length and encompass groups of genes with diverse functions, which can be linked to factors like virulence and metabolic diversity (Bespiatykh et al., 2021, Bottai et al., 2020). Additionally, the *IS6110* element has a notable impact on the genetic diversity of the MTBC. Transposition of *IS6110* into various coding or regulatory regions, and genomic deletions through homologous recombination between proximal copies, can also translate into significant genotypic variation at strain level (Alonso et al., 2013, Antoine et al., 2021). These mechanisms can all contribute to a varying repertoire of genetic features, much of which remains underexplored.

The pangenome represents the set of all genes present in a species. It comprises both the core genome, which consists of genes shared by all members of the species, and the accessory genome, which includes genes present in some but not all members. The accessory genome is often split into shell (present in most but not all genomes) and cloud (present in only a few genomes). Pangenomes are further classified as either open (each new genome adds a significant number of new genes to the population's accessory genome) or closed (newly analysed genomes do not significantly increase the accessory genome size) (Brockhurst et al., 2019 [↗](#)).

Previous research has focused on pangenomic investigations of particular MTBC lineages (Baena et al., 2023 [↗](#), Ceres et al., 2022 [↗](#), Sanoussi et al., 2021 [↗](#)) and a recent study investigated pangenome of most MTBC lineages using protein sequences (Silva-Pereira et al., 2024 [↗](#)). However, there is a gap in comprehensive pangenomic analyses encompassing both coding and non-coding regions. Despite the name, prokaryotic pangenome analyses tend to not focus on full genome comparisons, but rather on protein-coding sequences. This focus neglects non-coding RNAs and pseudogenes as well as essential elements in intergenic regions, such as promoters, terminators, and regulatory binding sites, despite their demonstrated role in selection and phenotypic impact (Tonkin-Hill et al., 2023 [↗](#)). Moreover, the inference drawn from pangenomic analyses is often influenced by short-read datasets. It has been shown that using short-read datasets can artificially inflate the size of the pangenome estimates, leading to incorrect assumptions about evolutionary scenarios (Marin et al., 2024 [↗](#)). Here, we used a diverse collection of closed MTBC genomes including human and animal-associated lineages to develop our understanding of MTBC pangenome and evolution incorporating both coding and non-coding regions. The objectives of this study are: (i) to establish the true size of MTBC pangenome using a highly curated dataset, (ii) to assess specific biological functions that are prevalent in the core and accessory genome and (iii) to assess gene content variation at both lineage and sub-lineage levels. By focusing on sub-lineages, we aim to provide a detailed characterisation of genetic diversity and structural differences that influence MTBC evolution and functional traits long term. We find that the MTBC has a small, closed pangenome and genome reduction via RDs is the driving evolutionary force that shapes the genomic repertoire of the species. We identified distinct genomic features among lineages and sub-lineages, including sub-lineage specific RDs, that may account for variations in virulence, metabolism, and antibiotic resistance.

Methods

Summary of sequencing data and quality control

Previous studies have shown that closed genomes should be used for the best estimation of pangenome size (Marin et al., 2024 [↗](#)). From a combination of previously published research and publicly available data, 328 non-duplicated, complete (single contig) and closed (as listed on NCBI) genome assemblies were selected for having (i) long-read sequencing only (Nanopore or PacBio) or (ii) hybrid sequencing with Illumina across the MTBC lineages. Information related to the geographical location, source of isolation, assembly accessions and quality metrics can be found in Supplementary Table S1. The quality of all assemblies was assessed using BUSCO v5.8.2 (Manni et al., 2021 [↗](#)) and only assemblies with $\geq 95\%$ completeness were used for further analysis. For parity, all assemblies were set to start on *dnaA* gene using the fixstart command of Circlator v1.5.5 (Hunt et al., 2015 [↗](#)). Complete circular assemblies were annotated using PGAP v6.4 with default options (Tatusova et al., 2016 [↗](#)). This approach is recommended by previous analyses (Marin et al., 2024 [↗](#)) as PGAP tends to annotate pseudogenes in cases of pseudogenisation and frameshift, rather than annotating them as new coding sequences. Visualisations of the PGAP annotation were made using the DNA Features Viewer Python library (Zulkower and Rosser, 2020 [↗](#)). Lineage classification was performed using TB-profiler v6.5 with the TBDB v.72ef6fa database (Phelan et

al., 2019) except for *M. tuberculosis* var *microti* (hereafter named *M. microti*) and *M. tuberculosis* var *pinnipedii* (hereafter named *M. pinnipedii*) which are not classified in this database and were assigned using spoligotype using SpolPred2 as implemented in TB-profiler (Napier et al., 2023).

Nanopore sequencing and assembly

We sequenced lineages and sub-lineages that either had no sequencing data available in public databases or were under-represented in the collection. These isolates belonged to L5 (n=2, L5.1 and L5.2), L4.6 (n=3, L4.6.1, L4.6.1.1 and L4.6.1.2), L4.7 (n=3), L9 (n=1), La2 (*M. caprae*, n=1) and *M. pinnipedii* (n=1). A list of genomes sequenced in this study and their assembly information can be found in Supplementary Table S1. Raw reads and assembled genomes were deposited in NCBI under the Bioproject PRJNA1085078. DNA extraction suitable for long-read sequencing was performed using the Genomic-tip 100/G (Qiagen Inc) as described previously (Ngabonziza et al., 2020). DNA concentration was quantified using a Qubit fluorometer and 1X dsDNA High Sensitivity (HS) assay kit (ThermoFisher Scientific). Nanopore sequencing was performed using the native barcoding kit 24 V14 (SQK-NBD114.24) according to the manufacturer's instruction on the MinION Mk1C platform with R10.4.1 flow cells. Basecalling was undertaken using Guppy v 6.5.7 with the super accurate (SUP) model. Sequence adaptors were trimmed using Porechop v0.2.4 with "--end_threshold 95" and "--middle_threshold 85" options (<https://github.com/rrwick/Porechop.git>). Sequences were assembled using flye v2.9.2 with default options (Kolmogorov et al., 2019) and iteratively polished using Racon v1.5.0 and the "consensus" subcommand of Medaka v1.8.0 (<https://github.com/nanoporetech/medaka.git>). Complete assemblies underwent Circlator, BUSCO and PGAP steps as above.

Pangenome analysis (protein coding genes)

Panaroo v 1.5.1 inferred the MTBC protein-coding pangenome (Tonkin-Hill et al., 2020) with and without the "--merge_paralogs" option to investigate gene duplication (paralogy). Additionally, "--clean-mode strict" and "--core_threshold 0.95" were applied to reduce potential contamination by removing genes found at low frequencies (present in fewer than 5% of genomes) and to define core alignment gene set for the subsequent phylogenetics, respectively. Ceres et al. (2022) showed that genes could be over-split by annotation software, creating additional accessory genes, which has been confirmed elsewhere (Marin et al., 2024). To counteract this we followed the approach recommended in Ceres et al. (2022) by merging genes that map to the same gene in a reference genome. The reference pangenome created by Panaroo was compared to the H37Rv genome (standard reference genome for the MTBC) using BLASTn with a similarity e-value cutoff of 1×10^{-10} (Zhang et al., 2000). The BLASTn result was then filtered to only include queries with $\geq 90\%$ identity and $\geq 75\%$ of the H37Rv gene length covered using in-house Python scripts (<https://github.com/conmeehan/pathophy>). A core genome alignment was constructed using the built-in MAFFT aligner included in Panaroo (Katoh et al., 2002). The core phylogeny was created using RAxML-NG v1.2.2 with a GTR substitution model (Kozlov et al., 2019). Tree visualisations were created using the Treeio package v1.20.0 (Wang et al., 2020).

Pangenome graph construction

The above pangenome analysis only compares coding gene content and omits all non-coding genes and intra-gene genetic content. To examine the total pangenome, including both coding and non-coding regions, all genomes were aligned into a graph using Pangraph v0.7.2 (Noll et al., 2023). A minimum genome region block size of 250bp was selected to look only at large polymorphisms. As above, an additional check was undertaken on blocks labelled as absent, to ensure they truly are not present in the given genome, and not just mis-labelled as absent by the Pangraph pipeline. BLASTn was used to compare the consensus sequence for that region and the genome it is predicted to be absent from. Any BLASTn results with an e-value less than $1e^{-30}$ and additionally covering 50% of the region length and a 50% identical match were re-labelled as present. Genomic regions were classified as a core region if present in all genomes or an accessory region if present

in some but not all genomes. The DNA sequences from genomic blocks present in at least one sub-lineage but completely absent in others were extracted to look for long-term evolution patterns in the pangenome. Known Regions of Differences (RDs) were identified using RDscan (Bespiatykh et al., 2021). Structural variants with Rv (H37Rv gene) loci overlapping previously annotated RDs were assigned the same RD names. Novel structural variants that did not align with any previously reported RDs were classified as new RDs if they are present in H37Rv, or as a new LSP if absent. Accessory regions were further classified as a deletion if present in over 50% of the sub-lineages or an insertion/duplication if present in less than 50% of sub-lineages. This approach provides a reference-free perspective for investigating gene content differences and labels relevant LSPs as reductive (deletions) or increasing (insertions/duplications).

Functional genome analysis

The reference pangenome created by Panaroo was translated using transeq v5.0.0 (Rice et al., 2000) to obtain the protein sequences. COG functional categories were assigned using eggNOG Mapper v5.0.2 using default options (Huerta-Cepas et al., 2017). Genes that did not return a match within the eggNOG database were included in the “not found” category. Functional domain annotation was performed with InterProScan v5.63-95 (Jones et al., 2014). For the gene association analysis, the accessory genome (genes present in less than 95% of genomes) was used as input for Coinfinder v1.2.1 (Whelan et al., 2020). The STRING database (<https://string-db.org/>) was used to infer interaction networks of genes. Predicted functional partners with a high confidence interaction score (≥ 0.7) by STRING v12.0 were considered in this study.

Statistical analysis

Statistical analysis was conducted using R studio v2024.04.2+764 (R Core Team, 2023). Due to the use of multiple public genomes created using different sequencing technologies, we looked for significant associations between accessory genome size and sequencing technology. For comparative purposes and to ensure sufficient sample size per group, all types of Illumina sequencing (e.g., MiSeq, HiSeq, etc) were combined, resulting in comparisons between different long-read technologies, with or without Illumina hybrid sequencing. Any sequencing technology group with fewer than three genomes was excluded. The Shapiro-Wilk test was used to test for normality and then a Kruskal-Wallis test was used to look for associations between accessory genome count (the number of missing genes in a genome) and sequencing platform(s). A Dunn’s test was then used to look for pairwise significant differences with a Bonferroni correction for multiple comparisons. The same process was undertaken on pseudogene counts, based on the GFF files output from PGAP.

Principal component analysis (PCA) based on gene presence and absence data from Panaroo-derived accessory genome contents and Pangraph-derived accessory region contents was performed using the prcomp package in R. The fviz_contrib function in Factoextra (v1.0.7) was used to identify the contribution of important variables from the PCA results (Kassambara and Mundt, 2020).

To assess the openness and pairwise uniqueness of the pangenome, micropan was used to estimate Heaps law, genome fluidity and rarefaction curves with 100 permutations (Snipen and Liland, 2015). An alpha of 1 or lower in Heap’s law indicates an open genome; higher than 1 indicates a closed genome.

Results

Overview of the genomic dataset

We combined 339 whole-genome sequences covering all known MTBC lineages except L10 for which no closed genome is available (Guyeux et al., 2024 [DOI](#)). This included 328 genomes published previously, as well as 11 genomes sequenced here from sub-lineages with few or no representatives in the public dataset to better cover the known phylogeny of MTBC (Supplementary Table S1). These include the first closed genomes for L5.2, L9, La2 (formerly *M. caprae*) and *M. pinnipedii*. All genomes were sequenced to a depth greater than 30x coverage, as required for >99% sequence recovery (Sanderson et al., 2024 [DOI](#)).

Except for Antarctica, our collection includes genomes from every continent, and different host species including humans, cattle, voles and BCG vaccine strains (Supplementary Figure S1-A). Africa has the highest diversity of MTBC genomes of all continents, which is consistent with the fact that the continent is known to have the most diverse MTBC strains (Supplementary Figure S1-B) (Gagneux, 2018 [DOI](#)). Most L1 genomes have been collected from East Africa, South and Southeast Asia, where this lineage is geographically common. L3 genomes were mostly isolated from India, while L2 and L4 genomes have been collected from different continents, representing the global distribution of these lineages. L5 and L6 genomes were exclusively found in West Africa, where they contribute significantly to the overall burden of TB in this region (Coscolla et al., 2021 [DOI](#)).

Across the total collection, genomes contained on average 3,950 protein-coding genes (CDS) (range 3,579-4,018), of which 76 were predicted to be pseudogenes (range 64-93) (Supplementary Table 1). No significant association was found between the sequencing platform and pangenome size (Kruskal-Wallis test: p-value < 0.001). Pairwise comparisons also found no significant differences in pangenome size (Dunn's test: p-value < 0.001). However, a less stringent threshold (p-value < 0.05) indicated a difference in pangenome size between Nanopore/Illumina hybrid assemblies and PacBio SMRT (no hybrid assembly), with Nanopore hybrid assemblies showing lower accessory gene counts. No difference in pseudogene count was observed based on sequencing technology, suggesting this was not impacted either.

Pangenome characteristics and functional diversity

Panaroo estimated a protein-coding pangenome (with merged paralogs) consisting of 4,118 genes, including 3,638 core genes ($\geq 99\%$ of genomes) and 480 accessory genes including 180 soft core ($\geq 95\%$ and $< 99\%$ of genomes), 182 shell ($\geq 15\%$ and $< 95\%$ of genomes) and 118 cloud ($< 15\%$ of genomes) genes. After mapping all the genes in the pangenome to H37Rv to look for over-splitting of annotated genes, we identified 86 genes (2% of total genes) that were misclassified as accessory genes, including 9 out of 180 soft core, 12 out of 182 shell and 65 out of 118 cloud genes (Table 1 [DOI](#) and Supplementary Table S2). This approach reduced the pangenome size from 4,118 to 4,032 genes and the accessory genome from 480 to 394 genes. Incorporating duplicated genes (i.e. not merging paralogs in Panaroo) increased the pangenome size to 4,427 genes, including 3,635 core and 792 accessory genes (195 soft core, 200 shell and 397 cloud). After combining genes based on mapping to H37Rv to reduce annotation splitting bias, the (including paralogs) pangenome size decreased to 4,321 genes, including 3,627 core genes and the accessory genome was reduced to 694 (187 soft core, 186 shell and 321 cloud) (Table 1 [DOI](#) and Supplementary Table S2).

Heap's Law analysis resulted in an alpha of 2 for the merged paralog dataset and 1.548 for the unmerged dataset. This indicates a closed or near closed pangenome based on both datasets. Genome fluidity was estimated to be 0.0085 for the merged paralog dataset and 0.0088 for the unmerged paralog dataset. This indicates any two randomly selected genomes only differ by 0.85-0.88% of gene content. (Supplementary Figure S2).

All subsequent analyses based on Panaroo was undertaken using the merged paralog dataset, unless otherwise stated.

Method	Pangenome statistics	Raw estimates	Corrected estimates
Panaroo Merged paralogs	Total Genes	4,118	4,032
	Core genes	3,638	3,627
	Accessory genes:	480	394
Panaroo Including paralogs	Total Genes	4,427	4,321
	Core genes:	3,635	3,627
	Accessory genes:	792	694

Table 1

Pangenome (Panaroo) size estimations with and without corrections and merged paralogs.

Corrected estimates are derived from filtering out false accessory genes through merging of genes with strong hits to the same gene in H37Rv ($\geq 90\%$ identity and $\geq 75\%$ gene length coverage).

To examine genomic diversity among lineages, we investigated variations in the size of the accessory genome among different lineages. This analysis revealed significant differences in accessory genome size among different lineages (Kruskal-Wallis test, $p = 2.2 \times 10^{-16}$). Post-hoc Dunn tests with Bonferroni correction identified significant pairwise differences between lineages (Supplementary Table S3). L6, La1 and *M. microti* have significantly smaller number of accessory genes per strain compared to others (**Figure 1**, Supplementary Table S3); L2 has a significantly smaller accessory genome in comparison with L1, L3, and L4.

Functional annotation of core and accessory genes showed that genes of unknown function were the largest functional group in both, with over 20% (**Figure 2**). The core genome of MTBC encodes a high proportion of genes involved in lipid transport and metabolism (category I), transcription (category K) and cellular transport and secretion systems (category N), indicating the importance of these biological functions in defining the species. Functional breakdown of the accessory genome also revealed variations in genes involved in DNA replication, repair and mobile elements (category L) and lipid transport and metabolism (category I) across the species. The functional composition of core and accessory genes with the inclusion of paralogs is shown in Supplementary Figure S3.

Pangraph revealed a similar distribution of core and accessory regions (**Table 2**). In total, 1,338 regions were identified, with 1,015 (75.8%) present in all genomes. The accessory regions were distributed as follows: 129 (9.6%) in the soft core, 140 (10.4%) in the shell and 54 (4%) in the cloud. Regions ranged in length from 250bp (the set lower limit) to 54.6kb (mean: 3.2kb, median: 1kb). Some regions (2.3% of total pangenome regions) were found to be mislabelled as absent in certain genomes, which were corrected using BLAST. This resulted in small corrections to the pangenome category distributions, primarily with an increase to the core region and reduction in the cloud accessory region count (**Table 2**), similar to the pangenome corrections above. Blocks that had these spurious hits discovered by BLASTn ranged in size from 254bp to 5.4kb (mean: 815bp, median: 584bp), suggesting this is an issue mainly impacting smaller Pangraph blocks.

After corrections, accessory blocks made up ~23% of the pangenome. These accessory blocks ranged in size from 250bp to 9.8kb (mean: 993bp; median 629bp). This suggests most accessory blocks each encompass only one or a few genes each, with most larger regions (i.e. all above 10kb) being common to all the MTBC strains.

Heaps law based on the Pangraph block distribution resulted in an alpha of 2 and a genome fluidity of 0.041 (i.e. 4.1% uniqueness between any two genomes). This indicates a closed genome, similar to the Panaroo-based analyses, but with more unique blocks per genome when both coding and non-coding regions are considered compared to coding sequences alone (Supplementary Figure 2).

Population structure analysis

The population structure was assessed using SNPs in the core genome (**Figure 3A**) and PCA plots were constructed based on presence-absence patterns in the accessory genome determined with Panaroo (includes coding genes only) and the accessory regions derived from Pangraph (includes both coding and non-coding genomic regions) (**Figure 3B-D**; Supplementary figure S4). In the PCAs, genomes group together based on their lineage, indicating variations in accessory genome are mainly structured by lineage (**Figure 3B-D**). The first principal component (PC1) highlights the contrast between L2 from other members of MTBC. In all PCA plots, L2 formed a distinct group, defined primarily by the important variables RD105 and RD207 deletions. RD105 serves as a classic marker for all L2 genomes and RD207 deletion is found in all L2.2 sub-lineages; the single L2.1 genome is separated from the L2.2 cluster due to the absence of RD207. The second principal component (PC2) demonstrates the difference between modern human-associated lineages (L2, L3 and L4) from other lineages. Important variables driving this separation are the TbD1 gene deletion in modern lineages, and gene deletions associated with RD11 (prophage phiRv2) in L6, L7,

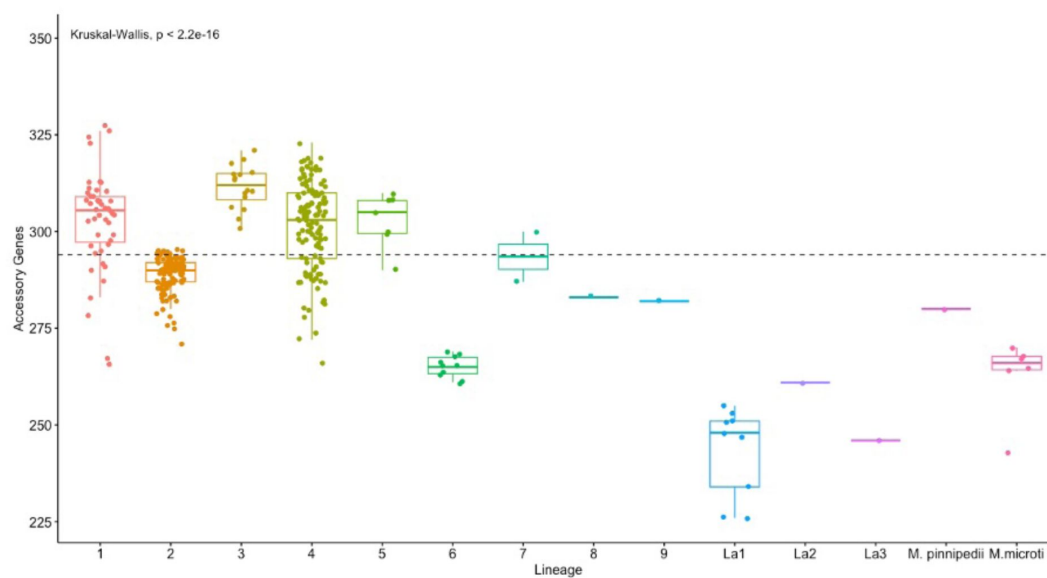


Figure 1

Boxplot showing the distribution of accessory genes within each lineage.

Lineages with a small number of genomes were excluded from the statistical analysis. L6, La1 and *M. microti* have significantly smaller accessory genomes compared to other lineages.

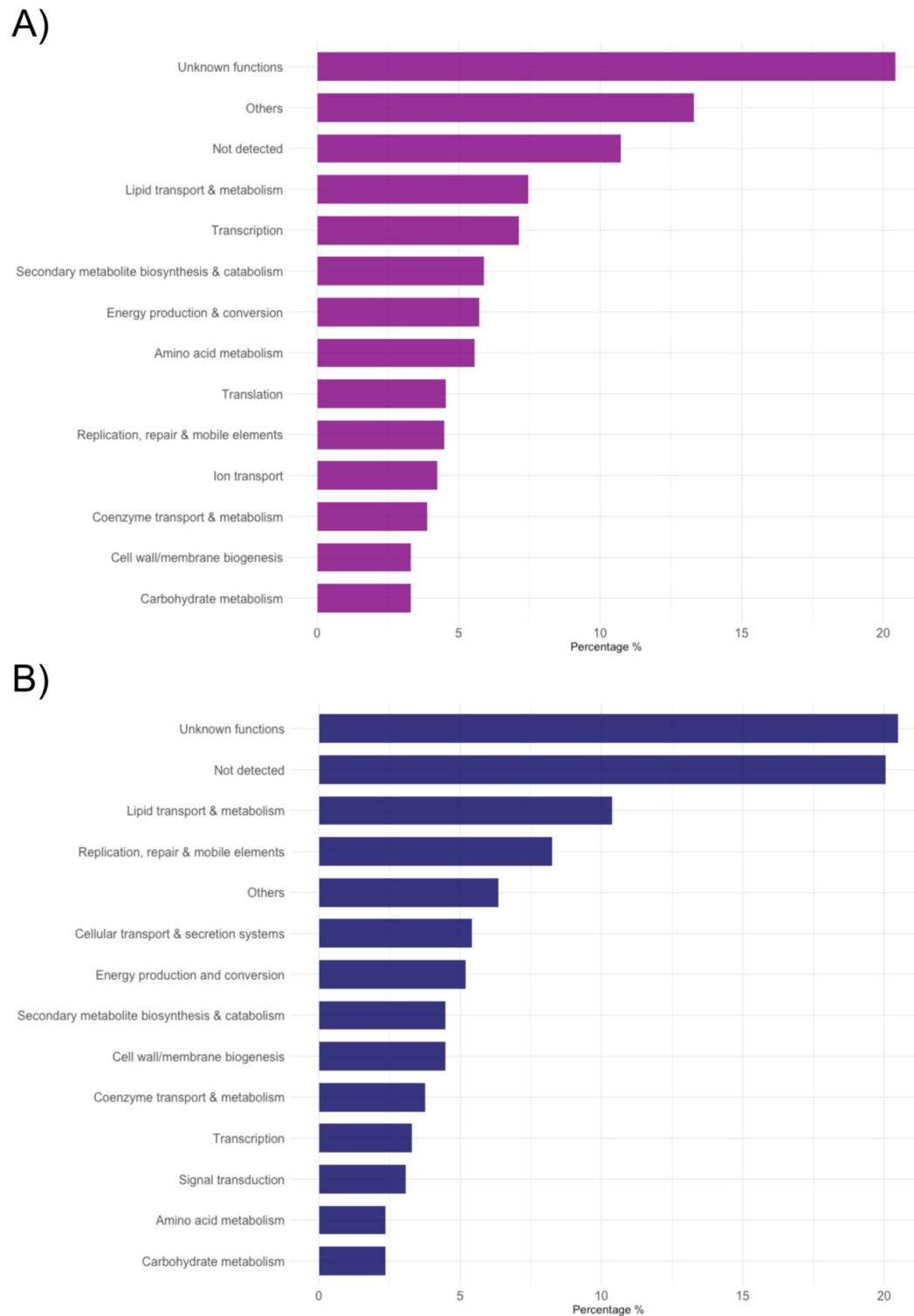


Figure 2

Analysis of the functional components in the core (A) and accessory (B) genome using EggNOG mapper and InterProScan.

Core and accessory genes are derived from Panaroo with merged paralogs. For without merged paralogs see Supplementary Figure S3.

Pangenome partition	Raw estimates	Corrected estimates
Total Regions	1,338	1,338
Core Regions	1,015	1,040
Soft core	129	124
Shell	140	143
Cloud	54	31

Table 2

Pangenome graph (Pangraph) size estimations with and without corrections.

Corrected estimates are derived from filtering out false labelling of absent regions using BLASTn.

La1 and La3 genomes and some L1 sub-lineages. L1 genomes exhibit considerable genetic diversity primarily due to RD3 (related to prophage phiRv1) and RD11. The latter was identified in all L1.1 sub-lineages and certain L1.2 sub-lineages (L1.2.1, L1.2.2, and L1.2.2.2). The positioning of the single L8 genome, (positive sign, **Figure 3B-D**) sits separate from all other lineages, likely due to being the most basal known lineage (Ngabonziza et al., 2020).

The accessory genome of MTBC sub-lineages

To counteract the potentially misleading effect of mis-annotations of some genomes on accessory genome analyses, we looked for larger patterns of genomic differences in the MTBC, shared by multiple genomes. The accessory genome (Panaroo) and accessory region (Pangraph) patterns outlined above were filtered to look only at those genes and larger genomic regions present in all strains of one or more sub-lineages and completely absent in others (i.e. sub-lineage-specific genes and LSPs).

Among the 394 filtered accessory genes, 111 were identified as specific to certain lineages and sub-lineages (Supplementary Figure S5-A, Supplementary Table S4). Four of these genes were absent from La3 but present in at least two copies in all other lineages: Rv3117 (cysA3) and Rv0815c (cysA2), involved in the formation of thiosulfate, and Rv3118 (sseC1) and Rv0814c (sseC2), involved in sulphur metabolism. Panaroo failed to identify partial gene deletions caused by RD9, RD10, RD13, RD105, RD207, and RD702. For example, RD702 is present (i.e. deleted) in all L6 and L9 genomes and covers a 641 bp region (**Figure 4**). RD702 affects less than a quarter of the *bglS* gene (470 bp from 2,076 bp) and *mymT* gene (126 bp from 162 bp gene) in the 5' to 3' direction as well as a regulatory RNA (ncRv0186c) in the 3' to 5' direction. The Pangraph approach identified most partial gene deletion and non-coding regions of the DNA that were impacted by genomic deletion (relative to H37Rv). However, the graph approach failed to identify four structural variants, including a gene deletion corresponding to RD8 (Rv3618) in L6 and the animal-associated genomes.

Gene association analysis using coinfinder confirmed that most accessory genes (as determined by Panaroo) co-occur in pairs or groups in larger multi-gene RDs (as determined by Pangraph) (Supplementary Figure S5). Thus, the small accessory genome is contained within an even smaller number of RDs. Coincident genomic blocks that were associated with lineages were RD8, RD9, RD10, and RD178, whereas those that were distributed across the phylogeny belonged to RD3 and RD11.

The filtered Pangraph results identified 82 lineage and sub-lineage-specific genomic regions ranging from 270 bp to 9.8kb, encompassing a total of 116 genes and 4 non-coding RNA (Supplementary Table S4). Many known lineage and sub-lineage-specific genomic regions were consistently identified using both methods (Panaroo and Pangraph) and found to be differentially present in two or more genomes, including TbD1, RD9, RD10, RD13, RD105, RD303, RD312, RD316, RD702, RD711, RD713, RD750, ND1, RDoryx, and N-RD25bov/cap (**Figure 4**).

Sub-lineage-specific accessory regions were classified as either RDs (deletions relative to H37Rv; present in most sub-lineages but absent in less than 50%) or LSPs (insertions/duplications; present in less than 50%). All sub-lineage-specific accessory regions were found to be RDs, i.e. present in H37Rv (L4.9 in **Figure 4**) and absent in only a minority of sub-lineages. Additional analysis of paralogous blocks, those present in all genomes but in multiple copies in some, did not reveal any new sub-lineage-specific accessory regions.

Overall, these findings point towards deletion-driven evolution of the MTBC.

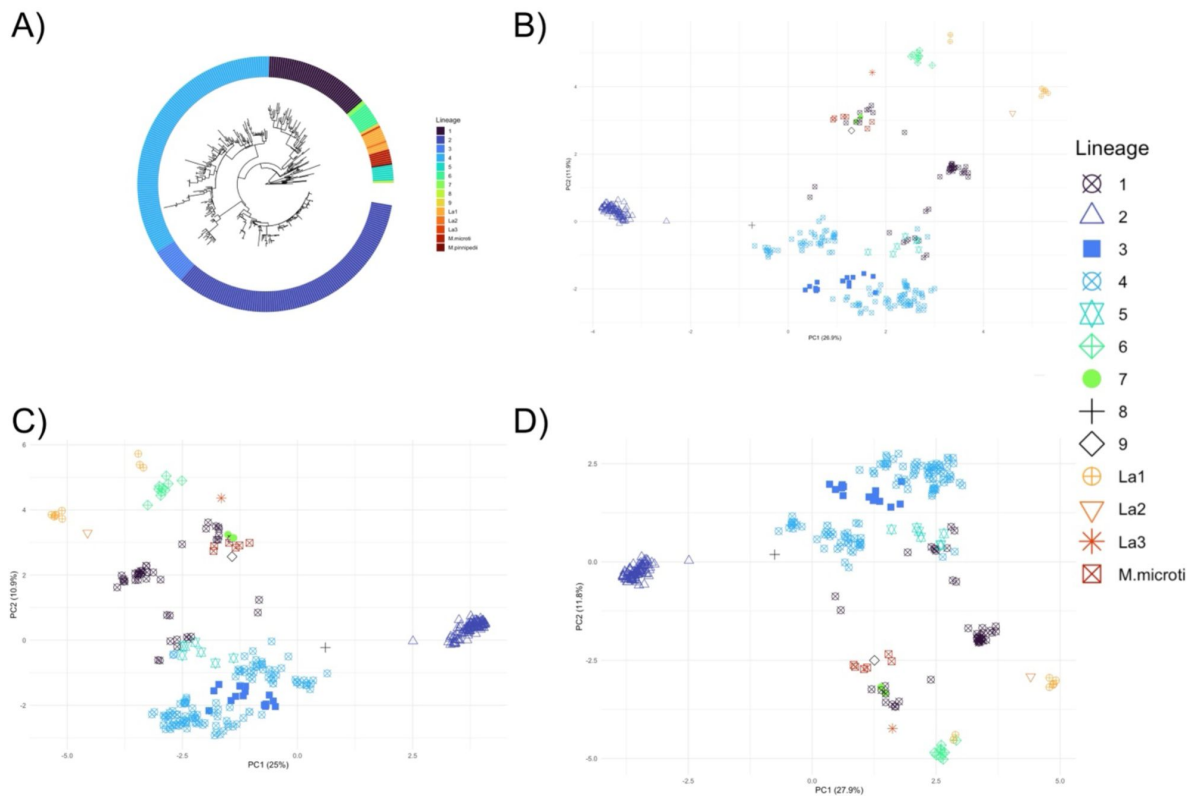


Figure 3

(A) phylogenetic tree based on MTBC core genome.

PCA based on the accessory genome data from (B) Panaroo (merged paralogs), (C) Panaroo (no merged paralogs) and (D) accessory regions data from Pangraph.

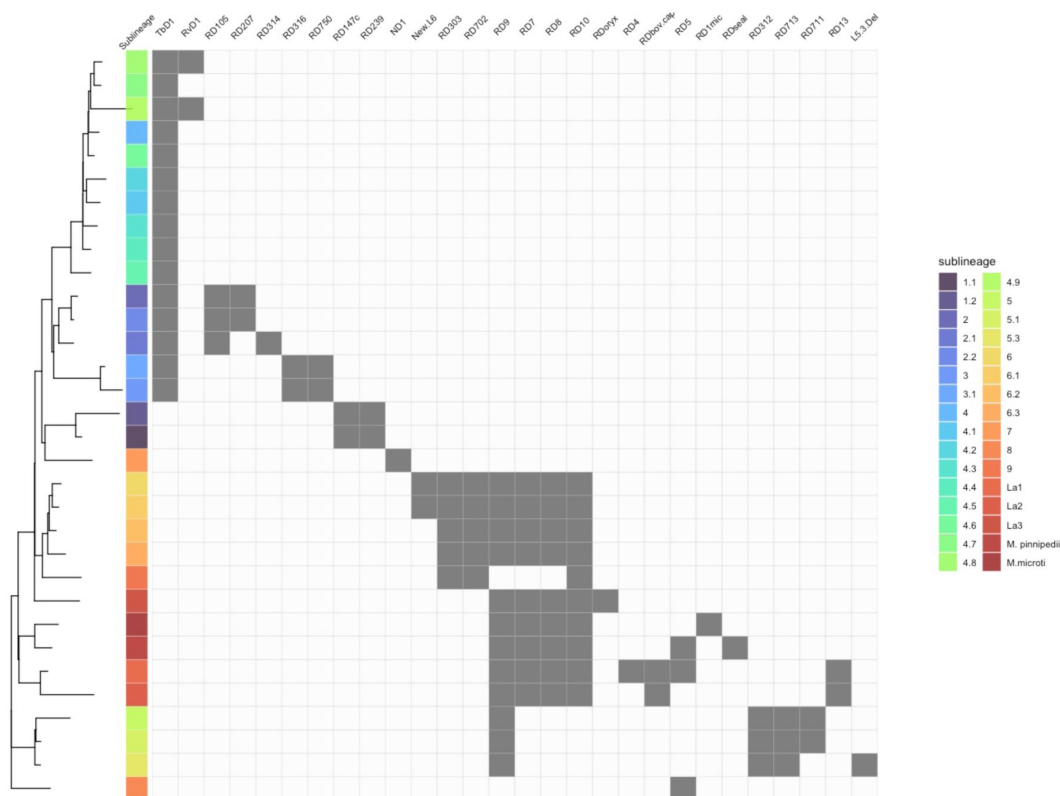


Figure 4

Sub-lineage specific regions of differences (RDs) identified using pangenome-based approaches.

Sub-lineages are shown on the Y-axis coloured as per the legend. H37Rv sits within L4.9 at the top of the Y-axis. RDs (structural variants present in all members of one or more sub-lineages and absent in all members of one or more other sub-lineages) are listed on the X-axis, grouped by their pattern of presence/absence. Only RDs detected by both the Panaroo and Pangraph-based approaches are shown. Grey boxes indicate that the region is absent from that sub-lineage. Only RDs that are due to divergent evolution and found present or absent in 2 or more genomes are shown here.

Discussion

The MTBC accessory genome is small and prone to inflation due to mis-annotation

The MTBC pangenome size has been reported to vary across studies. In a recent investigation conducted by [Marin \(2024\)](#), the accessory genome ranged from 314 to 2951 genes as a function of the assembly quality, software and specific parameters used. Genome sequencing using short-read technology has some limitations that can result in incomplete assemblies, particularly in regions with repetitive sequences that are difficult to sequence ([Meehan et al., 2019](#)). About 10% of the coding sequences of the MTBC genome are composed of repetitive polymorphic sequences known as the PE (Proline-Glutamic acid) and PPE (Proline-Proline-Glutamic acid) families, implicated in host-pathogen interactions and virulence ([Cole et al., 1998](#)). Incomplete assemblies can impact the accurate detection of genes and may lead to artificial inflation of the estimated pangenome size ([Ceres et al., 2022](#), [Marin et al., 2024](#)). We used a curated dataset to address these issues and estimate the true size of the MTBC pangenome. We showed that the MTBC has a compact accessory genome, totalling 10% of the total pangenome size (16% when including paralogs). This finding is consistent with recent research, which found a small accessory genome with a clonal genetic structure in the MTBC ([Baena et al., 2023](#), [Ceres et al., 2022](#), [Silva-Pereira et al., 2024](#)). Using a BLASTn analysis approach, we identified core genes that were designated as accessory and further reduced the accessory genome from 480 to 394 genes, a reduction of almost 20%. For example, two genes labelled as *lppA* and *group_1645*, both annotated as lipoproteins, were initially classified as accessory genes. However, these two genes exhibited 98% query coverage and 94% identity to the same gene in H37Rv, leading to their classification as a single core gene. Notably, 50% of the cloud genes (genes present in less than 15% of strains) were found to be core genes after this correction. These genes mapped to the reference genes using the BLAST approach and appeared to be fragmented genes caused by over-splitting in the PGAP annotation step.

The graph-based approach employed by Pangraph, encompassing all coding and non-coding regions, found a slightly larger accessory genomic content, with 22% of the total pangenome lying within accessory regions, after correction for spurious hits using BLASTn. This indicates that there is more variation to be uncovered when both coding and non-coding portions of the genome are assessed. However, correction of the Pangraph present/absent region patterns was required, showing caution and additional scrutiny is needed when assessing accessory region counts, especially for smaller blocks.

Identification of RDs was also likely impacted by the pangenome construction pipelines. Many partial RDs known to occur in specific lineages and sub-lineages were not identified by Panaroo. These results could be explained by the fact that Panaroo uses a relaxed alignment threshold along with neighbourhood gene information to cluster gene families ([Tonkin-Hill et al., 2020](#)). This can lead to the grouping of partially deleted genes with those that are complete. Pangraph appeared to overcome many of these issues as it uses the whole genome information as opposed to coding genes only. Despite these smaller limitations, both methods yielded consistent results indicating that the accessory genomic content (including coding and non-coding regions) of the MTBC is small, with much of the accessory genome contents shared between multiple strains (i.e. in the soft core and shell compartments). However, this does indicate that moving towards analysis of all genome regions, such as pangenome graphs, is required to identify the true accessory regions of difference within a population, especially if that population has a near closed pangenome.

The MTBC accessory genomic diversity is driven by reductive evolution

We employed a pangenomic approach to analyse the genetic diversity within the MTBC, aiming to understand its evolution and functional potential. In addition to inferring a gene-based pangenome with Panaroo, we also analysed variation in non-coding regions by inferring a pangenome graph with Pangraph. Examining non-coding regions is particularly important for species with limited genetic diversity, as they play a vital role in controlling gene expression, stress response and host adaptation (Arnvig et al., 2014 [↗](#)). Our study demonstrated that, although both methods produced reliable outcomes, the whole-genome alignment approach is needed to capture all genetic variations, especially those involving large structural variants (>200 bp), as much variation appeared to be outside of protein coding genes. Moreover, this method identified non-coding regions of the genome that were affected by genomic variation. Results from both methodologies revealed that the accessory genomic content of MTBC is a product of reductive evolution, that can be classified into divergent deletions arising from a single event (RDs) and convergent deletions, where the same deletion arises multiple times from independent events, with the former category being the more common. Coinfinder results also showed that coincident genes in the accessory genome primarily belonged to particular lineages.

However, a limitation of this analysis is that syntenic changes (those at the same places in different genomes) were not accounted for. Therefore, insertions or deletions which do not radically change the overall copy number but change the genome architecture could still occur and contribute to the overall evolutionary trajectory of the species. A more in-depth structural variation analysis accounting for such potential rearrangements is needed to assess the full impact of large insertions and deletions on the *M. tuberculosis* pangenome.

Regions of difference (RDs), which are large deletions often encompassing multiple genes, can act as distinctive markers not only for lineages, as demonstrated previously, but also for sub-lineages. These warrant further investigation to understand the biological consequences of such deletions on traits like virulence, drug resistance, and metabolism, along with SNP only analyses. On the other hand, we observed several genomic deletions that appear to be due to convergent evolution and may indicate gene instability (Brosch et al., 2002 [↗](#)). This includes RD3 and RD11 corresponding to the two prophages phiRv1 and phiRv2. One drawback of our study was that many of the lineages were poorly represented, especially in the L5, L6, L9 and all animal-associated lineages. The production of more closed genomes from the geographic regions where these are abundant (e.g. East and West Africa) will provide further information about the RD distributions within the MTBC and better understanding of the driving evolutionary forces.

All analysis undertaken here points towards genomic deletion as the main evolutionary factor creating the non-SNP-based variation within the MTBC. Reductive evolution is a driving factor of the other two *Mycobacterium* obligate pathogens, *M. ulcerans* and *M. leprae* (Stinear et al., 2007 [↗](#)). The *M. tuberculosis* genome has also been suggested to be a product of genome reduction as its genome is much smaller than its sister species' (Orgeur et al., 2024 [↗](#)). Our findings fit within this context, indicating that within-MTBC long-term evolution also tends to be reductive, with all discovered (sub-)lineage RDs being deletions relative to the complex as a whole. Much of this reduction appears to occur within the animal-associated lineages and the closely related human-associated lineages L5/L6/L9 (**figure 4** [↗](#)). This is not to say duplication is not also an important factor in driving evolution within this group. It has previously been shown that duplication of genes implicated in drug resistance and virulence genes (e.g. PE/PPE genes) are common within the accessory genome, albeit in a lineage-independent manner (Espinoza et al., 2025 [↗](#)). While accounting for duplicated genes did increase the accessory genome size here, especially the cloud

genome (Table 1), this did not translate into significantly different patterns between sub-lineages. More closed genomes of closely related strains are required to look for duplication-driven evolution in shorter timescales, especially if this is associated with changes in virulence.

The virulence potential of sub-lineages is differentially impacted by their accessory genomic content

The size of the accessory genome content varied significantly among different lineages. L2 genomes exhibited a high degree of clonality and a smaller accessory genome. L2 strains are characterised by a large number of *IS6110* copies, possibly due to the RD207, which affects the CRISPR-Cas type III-A system, responsible for adaptive immunity against mobile genetic elements (Alonso et al., 2013). While RD207 encompasses seven genes (Rv2814c-Rv2820c) (Bespiatykh et al., 2021), our analysis revealed that only two genes, Rv2819c and Rv2820c, are specifically absent in L2 and can serve as a genetic marker for this lineage. Other RDs affecting all L2 genomes are RD3 (9.2 kb), and RD105 (3.4 kb). The latter is exclusively present in all L2 genomes and previous research proposed a potential role of RD105 in conferring drug resistance in L2.2 strains (Qin et al., 2019). RD207 (7.4 kb), caused by the recombination of two *IS6110* elements, is also present in all L2 genomes except for L2.1. L3 genomes exhibited significantly less genomic deletion in comparison with L2, and RDs affecting all L3 genomes were RD316 (1.2 kb) and RD750 (765 bp). The genes affected by these RDs are involved in lipid metabolism (category I) and secondary metabolite biosynthesis and transport respectively (category Q), which may contribute to metabolic variations in L3 strains. L4 exhibits substantial genetic diversity primarily due to variations in the presence and absence of specific RDs at the sub-lineage level. For example, RD145 (4.7 kb) and RD182 (6.5 kb) were detected in L4.1.2.1 genomes isolated from Africa, Asia, Europe and South America. However, none of the affected genes appears advantageous or essential for MTBC growth and survival and this sub-lineage is successful worldwide (<https://tbd.r.lsh.tn.ac.uk/sra/browse>). Other L4 sub-lineage-specific RDs detected in this study with no known advantage are RD115 in L4.3.4, RD727 in L4.6.1, RD219 in L4.8, RvD1 in L4.8 and L4.9. RD4 in La1, L5.3 Del in L5.3, and RD750 in L3, RD147c in L1.

Nonetheless, some genomic deletions can affect the genes which confer growth advantages in various lineages. Disruption of these accessory genes has been shown to provide a fitness advantage for the *in vitro* growth of H37Rv by analysis of saturated Himar 1 transposon libraries (DeJesus et al., 2017). For example, RD174 specific to L4.3.4 intersects with RD743 specific to L5, and both deletions affect two genes (Rv1995 and Rv1996) including *cmtR* transcriptional regulator. The latter regulates bacterial stress response to metal deficiency and is important during persistence in phagocyte (Flores-Valdez et al., 2020). While the absence of these genes may not be necessary for *in vitro* growth (DeJesus et al., 2017) the impact of this deletion may be different in *in vivo* conditions. The low prevalence of L4.3.4 strains in the TB Profiler database may suggest either the lower fitness of this sub-lineage, or its presence in under-sampled geographic areas. RDs impacting the growth-advantage gene group were prevalent in L5 genomes (L5.3 Del, RD743 and RD711). For instance, RD711 affects the Rv1335 gene, which is annotated as a sulphur carrier protein involved in cysteine biosynthesis by PGAP. This may imply that the genome reduction observed in L5 has led to dependence on host factors for essential nutrients, consequently making these strains difficult to grow in a culture medium (Sanoussi et al., 2017).

L6 and animal-associated lineages accumulated a higher number of deletions including RD7, RD8, RD9 and RD10. Comparative analysis of the accessory genome size in L6, La1, and *M. microti* revealed significant differences compared to other lineages. The identified genomic regions in L6, L9, and animal-associated lineages impact genes associated with virulence, potentially explaining their reduced virulence in human hosts. RD7 contains the mammalian cell entry (*mce3*) operon implicated in the invasion of host cells, while RD9 includes genes related to biliverdin reduction, potentially aiding in protection against oxidative stress inside macrophages (Szklarczyk et al., 2023). RD10 impacts genes related to lipid storage crucial for MTBC transition to a dormant state

(Deb et al., 2009 [↗](#)). RD1BCG and RD1mic affect genes encoding the ESX-1 secretion system, crucial for virulence and phagosomal escape, with RD1BCG being deleted during BCG vaccine development (Szklarczyk et al., 2023 [↗](#)). Overall, the findings of this study suggest that, despite having a compact accessory genome, it can account for certain biological distinctions among MTBC. While genetic drift has been proposed as a key driver of diversity in MTBC (Hershberg et al., 2008 [↗](#)), recent research has argued that it does not fully explain the observed low genetic diversity in MTBC. Instead, purifying selection may be the dominant force, removing harmful mutations and maintaining genetic uniformity (Stritt and Gagneux, 2023 [↗](#)).

A need for more high-quality genomes and annotations

While this dataset represents high-quality MTBC genome sequences that cover most of its known diversity, it suffers from sampling bias as some lineages are over-represented whereas others have a few representatives. The over-representation of L2 and L4 in the dataset is primarily attributed to their association with Bioprojects analysing TB outbreaks or originating from countries with extensive sampling, highlighting a bias in sequencing efforts towards more prevalent or historically studied lineages. Many sub-Saharan African countries do not have a representative sample in our dataset, and this may explain the under-representation of L5 and L6 in the dataset. These lineages contribute significantly to the overall burden of TB in West Africa (Coscolla et al., 2021 [↗](#)). Additionally, we found that certain RDs that were previously assumed to be specific to certain lineages were identified in other lineages (Supplementary table S4). The observed variation in RD distribution emphasizes the need for an expanded genomic dataset to provide a more comprehensive understanding of their prevalence, specificity, and implications in the evolution of the MTBC. Future sequencing efforts should target high-burden TB countries, ensuring a comprehensive representation of the population structure to enhance our understanding of MTBC diversity and evolution. Closed genomes for closely related strains are also required to better understand short-term evolutionary patterns and the impact upon virulence within circulating strains. Additionally, investigating human transcriptomic data could further improve our understanding of the MTBC host adaptation, geographic distribution and pathogenicity.

Alongside these genomes, highly accurate annotation of the genes within is also needed. The availability of the H37Rv genome, which has undergone many rounds of intense reannotation (Modlin et al., 2021 [↗](#)), allowed for merging of genes that were over-split by PGAP. However, this meant only genes present in H37Rv could be re-merged where needed, potentially leaving other genes mis-annotated in the accessory genome. High quality, and high-throughout, annotation of other genomes will help rectify this in future expanded studies.

Conclusion

In conclusion, this study reveals that the MTBC has a small, near-closed pangenome, predominantly driven by genome reduction. This observation underlines the importance of genomic deletions in the evolution of the MTBC. We found many distinct genomic features across the MTBC, including sublineage-specific RDs, which likely play a crucial role in variations in virulence, metabolism, and antibiotic resistance. Finally, the study underscores the impact of non-coding regions and large structural variations on the MTBC's genetic diversity, offering a new perspective on MTBC pangenomics.

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Additional information

Author contributions

M.B. contributed to conceptualization, methodology, formal analysis, investigation, data curation, writing-original draft, writing – reviewing & editing, and visualization.

M.M. contributed to methodology, formal analysis, resources, data curation and writing – reviewing & editing.

D.W. contributed to methodology, formal analysis, data curation, writing – reviewing & editing.

M.F. contributed to methodology, data curation, writing – reviewing & editing and supervision.

J.C.T contributed to methodology, resources, and writing – reviewing & editing.

M.R.D.S. contributed to methodology, validation, formal analysis, investigation, writing – reviewing & editing and visualisation.

C.J.M. contributed to conceptualization, methodology, formal analysis, investigation, writing-original draft, writing – reviewing & editing, supervision, project administration and funding acquisition.

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Additional files

Supplementary Figure S1. The geography of the dataset. (A) sample collection distribution by country; (B) the number of genomes of each lineage included from each continent. [↗](#)

Supplementary Figure S2. Pangenome openness assessment using Heap's Law analysis, genome fluidity and rarefaction curve for A) Panaroo with merged paralogs; B) Panaroo with unmerged paralogs and C) Pangraph blocks. [↗](#)

Supplementary Figure S3. Analysis of the functional components in the core (A) and accessory (B) genome using EggNOG mapper and InterProScan. Core and accessory genes are derived from Panaroo without merged paralogs. [↗](#)

Supplementary Figure S4. MTBC accessory genome distribution based on (A) Panaroo (merged paralogs) and (B) Pangraph. The MTBC phylogenetic tree is shown on the left beside a coloured bar indicating the sub-lineage of each tip genome. The accessory genes/regions are indicated by columns in the heatmap with a blue box if present in that strain's genome. [↗](#)

Supplementary Figure S5. Coinfinder gene association heatmap. Coinfinder reveals gene association patterns within the accessory genome of the MTBC. Accessory genes, listed on the X-axis, cluster into groups (shown here by varying colours of the blocks). [↗](#)

Supplementary Figure S6. MTBC accessory genome distribution based on (A) Panaroo (merged paralogs) and (B) Pangraph. The MTBC phylogenetic tree is shown on the left beside a coloured bar indicating the sub-lineage of each tip genome. The accessory genes/regions are indicated by columns in the heatmap with a blue box if present in that strain's genome. [↗](#)

Supplementary Table S1. An overview of the genome dataset, both publicly acquired data and those strains sequenced within this study including assembly accessions and online locations, where relevant. BUSCO completeness information for each genome is also provided, as is the CDS and pseudogene count as predicted by PGAP. [↗](#)

Supplementary Table S2. A list of all gene groups which were combined to reduce over-splitting of pangenome due to annotation errors. Each gene groups name is listed alongside its original classification (Core, Sort-core, Cloud, Shell) and its new classification after merging with others after annotation correction. This is shown for both the pangenome with and without merged paralog setting enabled. [↗](#)

Supplementary Table S3. Pairwise comparisons of accessory genome size between lineages. A * indicates a significant difference between these lineages. Comparisons on both merged and unmerged paralog datasets are shown. [↗](#)

Supplementary Table S4. A list of all RDs (known and new) along with the genes contained in each and the lineages the regions are absent from. [↗](#)

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Reviewer #1 (Public review):

Summary:

In this paper, Behruznia and colleagues use long-read sequencing data for 339 strains of the *Mycobacterium tuberculosis* complex to study genome evolution in this clonal bacterial pathogen. They use both a "classical" pangenome approach that looks at the presence and absence of genes, and a pangenome graph based on whole genomes in order to investigate structural variants in non-coding regions. The comparison of the two approaches is informative and shows that much is missed when focusing only on genes. The two main biological results of the study are that 1) the MTBC has a small pangenome with few accessory genes, and that 2) pangenome evolution is driven by genome reduction. The second result is still questionable because it relies on a method that disregards paralogs.

Strengths:

The authors put together the so-far largest data set of long-read assemblies representing most lineages of the *Mycobacterium tuberculosis* context, and covering a large geographic area. They sequenced and assembled genomes for strains of *M. pinnipedi*, L9, and La2, for which no high-quality assemblies were available previously. State-of-the-art methods are used to analyze gene presence-absence polymorphisms (Panaroo) and to construct a pangenome graph (PanGraph). Additional analysis steps are performed to address known problems with misannotated or misassembled genes.

Weaknesses:

The main criticism regarding the dominance of genome reduction remains after two rounds of revisions. A method that systematically excludes paralogs is hardly suitable to draw conclusions about the relative importance of insertions/duplications and deletions in a clonal organism, where any insertion/duplication will result in a paralog. I understand that a re-analysis of the data might not be practical, and the authors have added a few sentences in the discussion that touch on this problem. However, the statements regarding the dominance of genome reduction remain too assertive given this basic flaw.

Here are the more detailed argument from the previous review:

In a fully clonal organism, any insertion/duplication will be an insertion/duplication of an existing sequence and thus produce a paralog. If I'm correctly understanding your methods section, paralogs are systematically excluded in the pangraph analysis. Genomic blocks are summarized at the sublineage level as follows (l.184): "The DNA sequences from genomic blocks present in at least one sub-lineage but completely absent in others were extracted to look for long-term evolution patterns in the pangenome." I presume this is done using blastn, as in other steps of the analysis.

So a sublineage-specific copy of IS6110 would be excluded here, because IS6110 is present somewhere in the genome in all sublineages. However, the appropriate category of comparison, at least for the discussion of genome reduction, is orthology rather than homology: is the same, orthologous copy of IS6110, at the same position in the genome, present or absent in other sublineages? The same considerations apply to potential sublineage-specific duplicates of PE, PPE, and Esx genes. These gene families play important roles in host-pathogen interactions, so I'd argue that the neglect of paralogs is not a finicky detail, but could be of broader biological relevance.

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Author response:

The following is the authors' response to the previous reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

In this paper, Behruznia and colleagues use long-read sequencing data for 339 strains of the Mycobacterium tuberculosis complex to study genome evolution in this clonal bacterial pathogen. They use both a "classical" pangenome approach that looks at the presence and absence of genes, and a pangenome graph based on whole genomes in order to investigate structural variants in non-coding regions. The comparison of the two approaches is informative and shows that much is missed when focussing only on genes. The two main biological results of the study are that 1) the MTBC has a small pangenome with few accessory genes, and that 2) pangenome evolution is driven by genome reduction. In the revised article, the description of the data set and the methods is much improved, and the comparison of the two pangenome approaches is more consistent. I still think, however, that the discussion of genome reduction suffers from a basic flaw, namely the failure to distinguish clearly between orthologs and homologs/paralogs.

Strengths:

The authors put together the so-far largest data set of long-read assemblies representing most lineages of the Mycobacterium tuberculosis context, and covering a large geographic area. They sequenced and assembled genomes for strains of M. pinnipedi, L9, and La2, for which no high-quality assemblies were available previously. State-of-the-art methods are used to analyze gene presence-absence polymorphisms (Panaroo) and to construct a pangenome graph (PanGraph). Additional analysis steps are performed to address known problems with misannotated or misassembled genes.

Weaknesses:

The revised manuscript has gained much clarity and consistency. One previous criticism, however, has in my opinion not been properly addressed. I think the problem boils down to not clearly distinguishing between orthologs and paralogs/homologs. As this problem affects a main conclusion - the prevalence of deletions over insertions in the MTBC - it should be addressed, if not through additional analyses, then at least in the discussion.

Insertions and deletions are now distinguished in the following way: "Accessory regions were further classified as a deletion if present in over 50% of the 192 sub-lineages or an insertion/duplication if present in less than 50% of sub-lineages." The outcome of this classification is suspicious: not a single accessory region was classified as an insertion/duplication. As a check of sanity, I'd expect at least some insertions of IS6110 to show up, which has produced lineage- or sublineage-specific insertions (Roychowdhury et al. 2015, Shitikov et al. 2019). Why, for example, wouldn't IS6110 insertions in the single L8 strain show up here?

In a fully clonal organism, any insertion/duplication will be an insertion/duplication of an existing sequence, and thus produce a paralog. If I'm correctly understanding your methods section, paralogs are systematically excluded in the pangraph analysis. Genomic blocks are summarized at the sublineage levels as follows (l.184): "The DNA sequences from genomic blocks present in at least one sub-lineage but completely absent in others were extracted to look for long-term evolution patterns in the pangenome." I presume this is done using blastn, as in other steps of the analysis.

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Reviewer #2 (Public review):

Summary:

The authors attempted to investigate the pangenome of MTBC by using a selection of state-of-the-art bioinformatic tools to analyse 324 complete and 11 new genomes representing all known lineages and sublineages. The aim of their work was to describe the total diversity of the MTBC and to investigate the driving evolutionary force. By using long read and hybrid approaches for genome assembly, an important attempt was made to understand why the MTBC pangenome size was reported to vary in size by previous reports. This study provides strong evidence that the MTBC pangenome is closed and that genome reduction is the main driver of this species evolution.

Strengths:

A stand-out feature of this work is the inclusion of non-coding regions as opposed to only coding regions which was a focus of previous papers and analyses which investigated the MTBC pangenome. A unique feature of this work is that it highlights sublineage-specific regions of difference (RDs) that was previously unknown. Another major strength is the utilisation of long-read whole genomes sequences, in combination with short-read sequences when available. It is known that using only short reads for genome assembly has several pitfalls. The parallel approach of utilizing both Panaroo and Pangraph for pangenomic reconstruction illuminated limitations of both tools while highlighting genomic features identified by both. This is important for any future work and perhaps alludes to the need for more MTBC-specific tools to be developed. Lastly, ample statistical support in the form of Heaps law and genome fluidity calculations for each pangenome to demonstrate that they are indeed closed.

Weaknesses:

There are no major weaknesses in the revised version of this manuscript.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

l. 27: "lineage-specific and -independent deletions": it is still not clear to me what a lineage-independent, or convergent, deletion is supposed to be. TBD1, for instance, is not lineage-specific, but it is also not convergent: it occurred once in the common ancestor of lineages 1, 2, and 3, while convergence implies multiple parallel occurrences.

We have changed this and in other places to more evolutionary terms, such as divergent (single event) and convergent (multiple events), or explain exactly what is meant where needed.

l. 118: "where relevant", what does that mean?

This was superfluous to the description and so is now removed.

l. 178ff.: It is not clear to me what issue is addressed by this correction of the pangenome graph. Also here there seems to be some confusion regarding orthologs and paralogs. A gene or IS copy can be present at one locus but absent at another, which is not a mistake of Pangraph that would require correction. It's rather the notion of "truly absent region" which is ambiguous.

We have changed the text to be more specific on the utility of this step. Since it is known that Panaroo mislabels some genes as being absent due to over splitting (see Ceres et al 2022 and our reclassification earlier in the paper), we wanted to see if the same occurred in Pangraph. We have modified the methods text to be more specific (line 181) and in the results included the percentage of total genes/regions affected by this correction.

In relation to copy number, Pangraph is not syntenic in its approach; if a region is present anywhere it is labelled as present in the genome. Pangraph will look for multiple copies of that region (e.g. an IS element) but indeed we did not look for specific syntenic changes across the genomes. This would be a great analysis and something we will consider in the future; we have indicated such in the discussion (line 454).

l. 305: "mislabelled as absent": see above, is this really 'mislabelled'?

See answer to question above

l. 372: "using the approach": something missing here.

This was superfluous to the description and so is now removed.

l. 381: the "additional analysis of paralogous blocks" (l. 381) seems to suffer from the same confusion of ortho- and paralogy described above: no new sub-lineage-specific accessory regions are found presumably because the analysis did consider any copy rather than orthologous copies.

Paralogous copies were looked for by Pangraph, and we did not find any sub-lineage where all members had additional copies compared to other sub-lineages. Indeed, single genomes could have these, and shorter timescales could see a lot of such insertions, but we looked at longer-scale (all genomes within a sub-lineage) patterns and did not find these. These limitations are already outlined in the discussion.

l. 415: see above. There is no diagnosis of a problem that would motivate a "correction". That's different from the correction of the Panaroo results, where fragmented annotations have been shown to be a problem.

Of interest, the refining of regions did re-label multiple regions as being core when Pangraph labelled it as absent from some genomes was at about the same rate as the correction to Pangraph (2% of genes/regions). This indicates there is a stringency issue with pangraph where blocks are mislabelled as absent. The underlying reason or this is not clear but the correction is evidently required in this version of Pangraph.

l. 430ff.: The issue of paralogy and that the "same" gene or region is defined in terms of homology rather than orthology should be addressed here. For me the given evidence does not support the claim that deletion is driving molecular evolution in the MTBC.

As outlined above, indeed paralogy may be driving some elements of the overall evolutionary patterns; our analysis just did not find this. Panaroo without merged paralogs did not find paralogous genes as a main differentiating factor for any sub-lineage. Pangraph also did not find multiple copies of blocks present in all genomes in a sub-lineage. As outlined above, indeed single genomes show such patterns but we did not include single genome analyses here, and outline that as a next steps in the discussion. We have also linked to a recent pangenome paper that showed duplication is present in the pangenome of Mtb, although not related to any specific lineage (Discussion line 485).

l. 443 ff: "lineage-independent deletions (convergent evolution)": see above, I still think this terminology is unclear

This has now been made clearer to be specifically about convergent and divergent evolutionary patterns.

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